Expanding Quantum Horizons: Monolithically Integrated C+L-Band Entangled Photon Source for Next-Gen Optical Networks

A BSc Thesis

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Abstract

Growing optical network capacity demands have spurred exploration of extending from the standard C-band (1530–1565 nm) to the adjacent L-band (1565–1625 nm) for a doubled capacity boost. Integrated sources generating entangled photons exhibit promise as platforms for quantum photonic information procedures due to their scalability and reliability. Furthermore, devices relying on electrical injection eliminate the necessity for optical alignment, streamlining experimental setups. This study presents the implementation and results of a versatile two-photon source that combines a C-band wavelength-selective switch (WSS) and an L-band dense wavelength-division multiplexer (DWDM), enabling comprehensive spectral routing in a single configuration. Our source employs a Bragg-reflection waveguide diode laser (BRL) configuration with direct intracavity phase-matching, driving spontaneous parametric down-conversion and yielding polarization-entangled photon pairs. The experimental results reveal a clear diagonal coincidence pattern consistent with the energy conservation condition of SPDC. This setup, tailored for flex-grid optical networks, showcases the potential for maximizing entanglement resource utilization across the entire C+L-band range.

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Chapter 1

Introduction

Quantum networks are of great importance due to their potential to revolutionize the field of information and communication technology. Multiplexing allows for the efficient utilization of resources and enhances the capabilities of these systems. With the capability of quantum networks increasing, the demands for increased optical network capacity are escalating. Thus, there is a growing interest in expanding beyond the conventional C-band (1530–1565 nm) to leverage the adjacent L-band (1565–1625 nm) as a means to double capacity.

Integrated sources capable of generating entangled photons have emerged as promising platforms for quantum photonic information processing, given their scalability and reliability. Additionally, devices reliant on electrical injection offer the advantage of obviating the need for intricate optical alignment procedures. While previous experiments have successfully demonstrated the generation of broadband polarization-entangled photon pairs for flexible-grid quantum networks [8], these approaches necessitate the injection of pump photons from external sources such as waveguides. This reliance on numerous optical components raises significant practical challenges for the implementation of large-scale optical networking protocols.

In this study, our primary objective is to develop an on-chip two-photon source that integrates a C-band wavelength-selective switch (WSS) and an L-band dense wavelength-division multiplexer (DWDM) for comprehensive spectral routing within a single configuration.

A Bragg-reflection waveguide diode laser (BRL) configuration with direct intracavity phase-matching will be used to drive spontaneous parametric down-conversion (SPDC), resulting in pairs of polarization-entangled photons, as previously demonstrated in our group [3]. The use of a BRL will eliminate the need for off-chip compensation or interferometry, simplifying the entangled photon production process.

These entangled photons from the BRL will then be directed towards a separate WSS and DWDM, each of which will allow specific wavelength bands to pass through. The output from the WSS and DWDM will be sent to two single photon detectors, allowing us to measure photon coincidence counts across a broad range of wavelength channels. We will conduct quantum state tomography on the system for various channels to assess the strength of entanglement over the entire bandwidth of the entangled photons.

When this research is conducted and we achieve our goals, we will not only optimize the utilization of entanglement resources across the entire C+L-band range but also reduce the form factor by utilizing a monolithic source of entangled photons. This source therefore has the potential to significantly benefit various quantum communication protocols, particularly those employing multiplexing. By achieving this, my research contributes to the advancement of quantum photonics and its practical applications in quantum information processing and secure communication.

Chapter 2

Background

2.1 Nonlinear Optical Processes

Nonlinear optical processes are phenomena in which the response of a material system to an applied optical field depends nonlinearly on the field strength [1].
In nonlinear optics, in general, the polarization $\tilde{P}(t)$ can be expressed as a power series in the field strength $\tilde{E}(t)$ as

$$\begin{aligned}\tilde{P}(t) &= \epsilon_0[\chi^{(1)}\tilde{E}(t) + \chi^{(2)}\tilde{E}^2(t) + \chi^{(3)}\tilde{E}^3(t) + ...] \\ &\equiv \tilde{P}^{(1)}(t) + \tilde{P}^{(2)}(t) + \tilde{P}^{(3)}(t) + ...\end{aligned}$$

where ϵ_0 is the permittivity of free space, $\chi^{(1)}$ is the linear susceptibility, and $\chi^{(2)}$ and $\chi^{(3)}$ are the second- and third-order non-linear optical susceptibilities, respectively.

2.1.1 Second Harmonic Generation

Second harmonic generation (SHG), shown in Figure 2.1, is a second-order nonlinear optical process in which an input source of frequency ω is injected into a nonlinear medium, which generates a second beam of frequency 2ω.
When a crystal with a nonzero second-order susceptibility $\chi^{(2)}$ is exposed to a laser beam with an electric field described as follows,

$$\tilde{E}(t) = Ee^{-i\omega t} + E^*e^{i\omega t}$$

the second-order polarization created in the crystal is

$$\tilde{P}^{(2)}(t) = 2\epsilon_0\chi^{(2)}EE^* + \epsilon_0\chi^{(2)}(E^2e^{-i2\omega t} + E^{*2}e^{i2\omega t})$$

where the second term leads to the generation of light at the second-harmonic frequency 2ω [1].

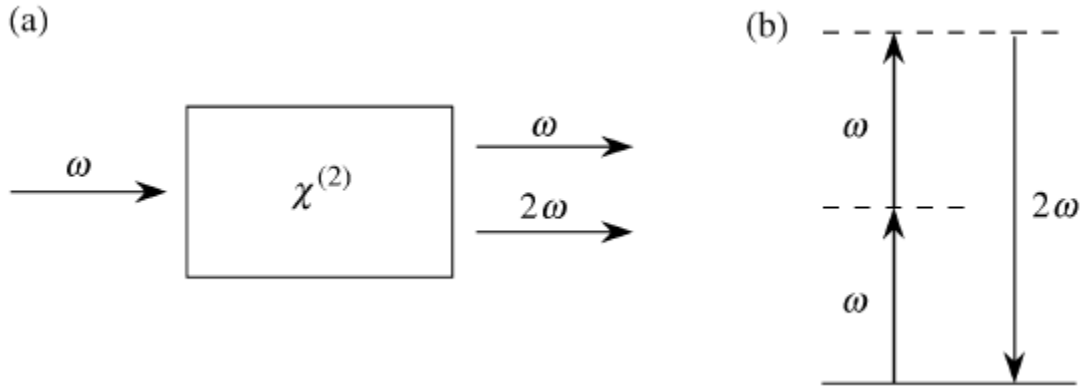


Figure 2.1: (a) Symbolic diagram of SHG in which photons of frequency 2ω are generated from an input beam of frequency w [1]. (b) An energy-level diagram that describes the SHG process.

2.1.2 Difference Frequency Generation

Difference frequency generation (DFG), shown in Figure 2.2, is a second-order nonlinear process in which two input sources of frequencies ω_1 and ω_2 are injected into a nonlinear medium, which generates a third beam of frequency $\omega_1 - \omega_2$.

Similar to SHG, when a crystal with a nonzero second-order susceptibility $\chi^{(2)}$ is exposed to a laser beam with an electric field described as follows,

$$\tilde{E}(t) = E_1e^{-i\omega_1 t} + E_2e^{-i\omega_2 t} + \text{c.c.}$$

(c.c. = complex conjugates) the second-order polarization created in the crystal is

$$\begin{aligned}\tilde{P}^{(2)}(t) =& 2\epsilon_0\chi^{(2)}[E_1E_1^* + E_2E_2^*] \\ &+ \epsilon_0\chi^{(2)}[E_1^2e^{-2i\omega_1 t} + E_2^2e^{-2i\omega_2 t} + 2E_1E_2e^{-2i(\omega_1+\omega_2)t} + 2E_1E_2^*e^{-2i(\omega_1-\omega_2)t} + \text{c.c.}]\end{aligned}$$

where the terms $2\epsilon_0\chi^{(2)}E_1E_2^*e^{\pm 2i(\omega_1-\omega_2)t}$ lead to the generation of light with a new

frequency $\omega_1 - \omega_2$.

The other terms in the above equation are related to other nonlinear processes, but usually only one of these processes occur [1].

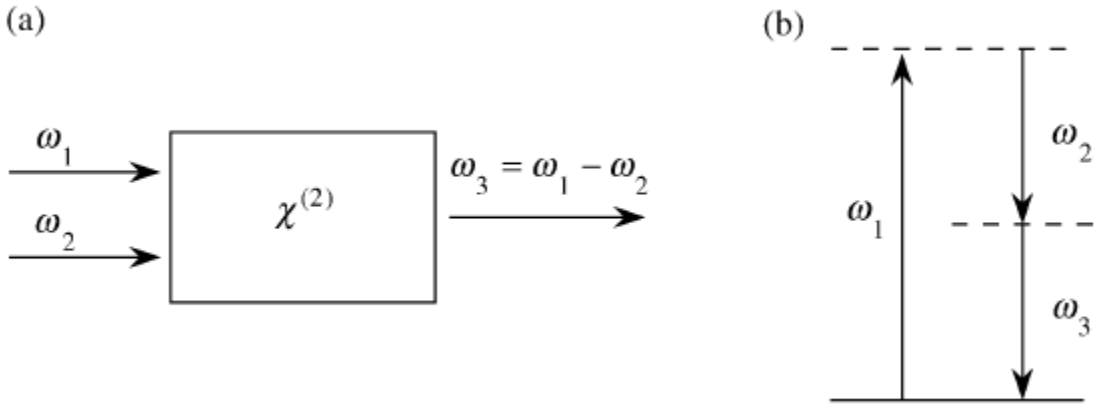


Figure 2.2: (a) Symbolic diagram of DFG in which photons of frequency ω_1 and ω_2 are inputted and generate an output photon of frequency $\omega_1 - \omega_2$ [1]. (b) An energy-level diagram that describes the DFG process.

2.1.3 Spontaneous Parametric Down-Conversion

Spontaneous parametric down-conversion (SPDC) is a second-order nonlinear optical process that involves an input photon, usually referred to as the pump photon, which after undergoing SPDC, splits into two photons called the signal and idler photons.

To achieve successful SPDC, two conditions need to be met; energy conservation and phasematching.

The Energy Conservation Condition

The energy of a photon with frequency ω is $E = \hbar\omega$ so to satisfy energy conservation, the frequencies of the photons must satisfy

$$\omega_p = \omega_s + \omega_i$$

where ω_p, ω_s, and ω_i are the frequencies of the pump, signal, and idler photons respectively.

The Phasematching Condition

The wave vector of a photon with frequency ω is given by $k = \omega n(\omega)/c$. Where $n(\omega)$ is the effective refractive index. The phasematching condition is thus

$$\Delta k = k_p - k_s - k_i = 0 \text{ or equivalently } \omega_p n(\omega_p) = \omega_s n(\omega_s) + \omega_i n(\omega_i)$$

where k_p, k_s, and k_i are the wave vectors of the pump, signal, and idler photons respectively.

Entanglement from SPDC

Type-0 and type-I phasematching are where the signal and idler have the same polarization (but have differing pump polarizations), and type-II phasematching is where the signal and idler photons have opposite polarizations. These phasematching conditions are not limited to SPDC and also apply to other second order nonlinear effects such as SHG.

Type-II SPDC can be used to generate entangled photon pairs, where the signal and idler photon pairs always have one extraordinary and one ordinary polarized photon. The output state is thus

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|H,V\rangle_{s,i} + e^{i\phi}\,|V,H\rangle_{s,i})$$

where $|H\rangle$ and $|V\rangle$ are the extraordinary and ordinary polarized states, s, i represent the signal and idler photons, and ϕ is the relative phase due to the birefringence [2]. It can be shown that this state is not separable in any basis, and is thus entangled.

2.2 Phasematching Techniques

Birefringence properties in a crystal can be exploited to achieve phasematching. In materials with significant difference between the refractive index of the ordinary orientation and the extraordinary orientation, phasematching is achieved by injecting photons with different polarizations. This method is known as birefringent phasematching [1].

In a potential such as an optical cavity, there are specified modes or eigenstates. The order of the mode specifies the probability distribution of a photon, and higher order modes have lower effective refractive indices [13].

In modal phasematching, the pump photons occupy a higher order mode than the signal and idler photons in order to achieve phasematching [1].

Quasi phasematching involves periodically switching the sign of the χ^2 coefficient so that any phase mismatch caused by one section is corrected by another section with opposite sign χ^2 coefficient, allowing for efficient nonlinear conversion even when the phase mismatch is high [1].

2.3 Semiconductor Lasers

A laser system comprises a gain medium enclosed within an optical cavity, and its operation involves the injection of energy through a pumping mechanism. In this process, electrons within the gain medium become excited through a pumping system, leading to spontaneous emission as they return to lower energy states, emitting photons. The optical cavity guides photons in the desired direction, facilitating stimulated emission where the emitted photons are coherent with the stimulating ones. Figure 2.3 shows a basic diagram of a laser.

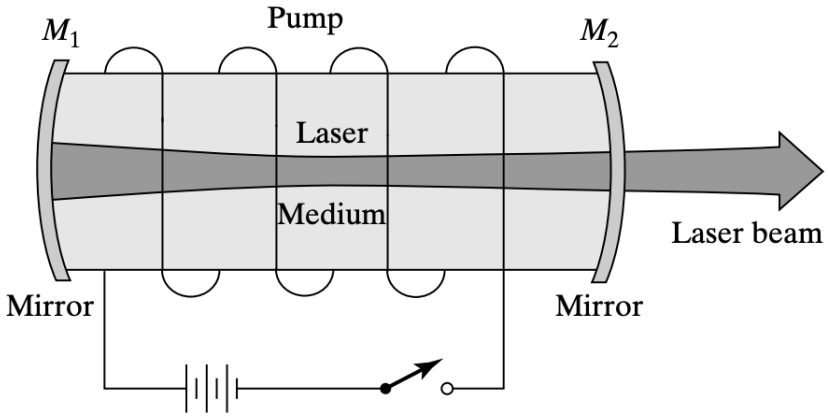


Figure 2.3: Basic diagram of a laser from [14].

Semiconductor lasers leverage a current source as the pumping mechanism. In this context, the injection of current into the semiconductor material induces recombinations between electron-hole pairs, generating photons through spontaneous emission. The guided photons within the optical cavity enable stimulated emission, resulting in the coherent amplification of the emitted light. This mechanism defines the fundamental operation of semiconductor lasers, where the use of current as a pumping source plays a pivotal role in achieving the desired optical output [14].

2.4 Bragg Reflection Lasers

2.4.1 Bragg Reflection Waveguides

The Bragg reflection waveguide (BRW) facilitates effective nonlinear conversion by exact modal phasematching. This involves guiding a higher-order mode referred to as the Bragg mode, as depicted in Figure 2.4, alongside the fundamental total internal reflection (TIR) mode.

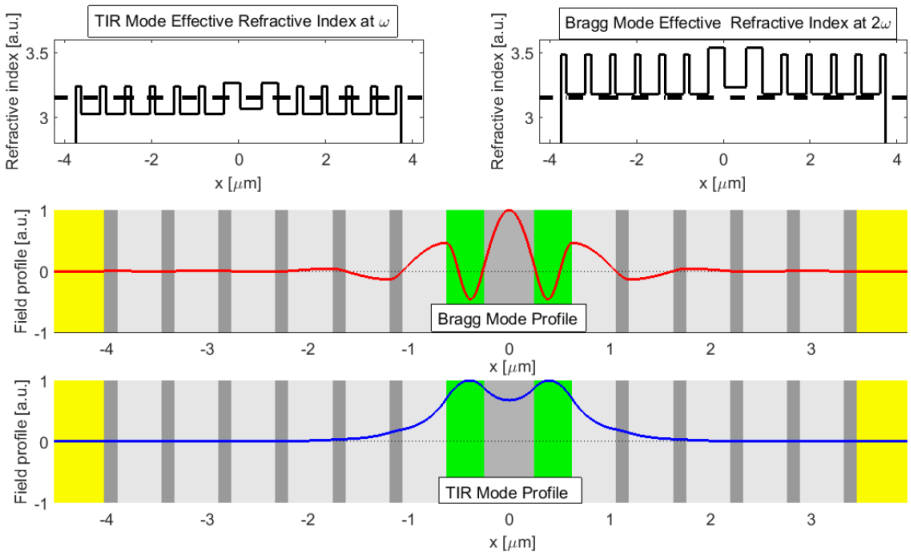


Figure 2.4: "Top left: Effective indicex of the TIR mode. Top right: Effective index of the Bragg mode. Middle: Bragg mode field profile for frequency 2ω. Bottom: TIR mode field profile for frequency ω." From MASc Thesis by Meng Lon Iu [4].

The BRW consists of layers comprising $Al_xGa_{1-x}As$ with different concentrations of x," forming a periodic arrangement of refractive indices. The structure of BRW involves two quarter-wave stacks separated by a guiding core layer. These quarter-wave stacks function as Bragg mirrors, reflecting and directing light along the core layer.

In the context of type-II SPDC, the signal and idler photons are found in the TE and TM TIR modes, while the pump photon is associated with the Bragg mode, as noted by [4]. The nonlinear capabilities of BRW have been verified in terms of performance, as illustrated in [5].

2.4.2 Bragg Reflection Waveguide Diode Lasers

The Bragg reflection waveguide diode laser (BRL), shown in Figure 2.5, is an active BRW that lases at the Bragg mode.

A p-n junction is established through the doping of the structure to create a diode junction. Electrical contacts can be applied to the structure to facilitate the injection of electrical carriers. To achieve population inversion, a stressed InAlGaAs quantum well is introduced into the core layer, leading the quantum well to operate as a gain medium. Subsequently, a basic semiconductor-air interface at the facet, with a reflectivity of 33 %, is sufficient to form the optical cavity.

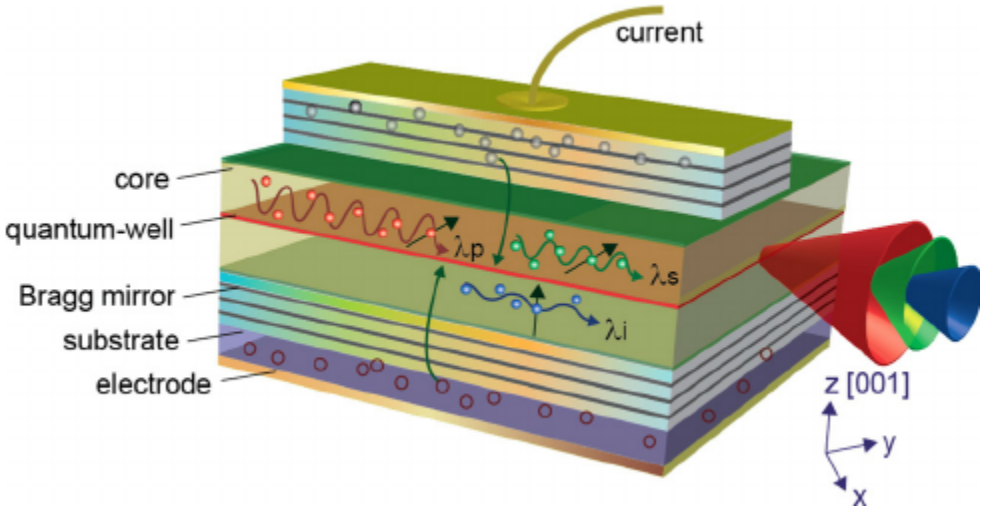


Figure 2.5: Bragg reflection waveguide diode laser from [15].

The BRL has been demonstrated in [6].

2.4.3 BRL Performance

Temperature Effect

Temperature has the ability to alter the bandgap of the quantum well, consequently influencing the lasing wavelength. A demonstrated correlation between the temperature of the BRL and the resulting lasing wavelength is established, as indicated in the study in [12].

With increasing temperature, the phasematching position – signifying the wavelength where nonlinear effects like SHG or SPDC occur – also experiences a linear increase. Notably, the phasematching and lasing wavelengths exhibit slightly different rates of change, enabling a dynamic tuning of the phasematching condition.

Current Effect

When a direct current is applied to the BRL, ohmic heating leads to an increase in temperature. Consequently, the lasing wavelength exhibits a linear proportionality to the injection current.

The impacts of temperature and current can be highly advantageous for experiments, offering a means to finely tune the BRL for lasing and phasematching at a specific wavelength of interest. The demonstrated effects of temperature and current tuning are detailed in [4].

2.5 Quantum State Tomography

Quantum state tomography, often referred to as photonic state tomography, is a powerful technique employed to characterize the quantum state of a photonic system. In the realm of quantum information processing and quantum communication, accurately determining the quantum state of individual or entangled photons is crucial for the implementation of reliable quantum protocols. Quantum state tomography provides a comprehensive means to reconstruct the complete density matrix that describes the quantum state, capturing both the amplitude and phase information associated with the photon's quantum properties.

The process of quantum state tomography involves performing a series of measurements on the quantum state of interest across different bases. These measurements are chosen strategically to extract sufficient information about the state's components. The obtained measurement outcomes are then processed using statistical methods and algorithms to reconstruct the full density matrix, which characterizes the quantum state in terms of its purity, entanglement, and other relevant parameters. Quantum state tomography plays a pivotal role in validating the success of quantum operations, providing a detailed and quantitative assessment of the generated quantum states. As quantum technologies advance, the precision and efficiency of quantum state tomography become increasingly crucial for ensuring the reliability and accuracy of quantum information processing tasks in various applications, including quantum communication and quantum computing [7].

2.6 Flex-Grid Quantum Networks

Quantum communication networks have emerged as transformative paradigms with the potential to revolutionize information and communication technology. The efficient utilization of resources and the enhancement of system capabilities are critical considerations for the development of robust quantum networks. The increasing capabilities of quantum networks have led to a surge in demand for enhanced optical network capacity, prompting a keen interest in extending beyond the conventional C-band (1530–1565 nm) to encompass the adjacent L-band (1565–1625 nm). This extension is envisioned as a strategic means to double the network's capacity, laying the groundwork for advanced quantum applications.

Broadband polarization-entangled source for C+L-band flex-grid quantum networks

One important contribution to this domain is the paper titled "Broadband polarization-entangled source for C+L-band flex-grid quantum networks" by Alshowkan et. al. [8]. This work addresses the imperative to optimize the utilization of entanglement resources across the C+L-band range. By introducing an ultrabroadband two-photon source designed for flex-grid optical networks, the paper employs flex-grid wavelength division multiplexing and wavelength-selective switches for comprehensive spectral routing. The integration of C+L-band wavelength-selective switches enables efficient utilization of the available spectrum, showcasing the potential for large-scale optical networking protocols.

The apparatus implemented in Alshowkan et. al. is shown in Figure 2.6. Here, a periodically poled lithium niobate (PPLN) waveguide produces polarization entangled photons via type-0 SPDC, in the bell state $|\Phi^+\rangle \propto |HH\rangle + |VV\rangle$. Wavelength selective switches are used for spectral routing.

Flexible entanglement-distribution network with an AlGaAs chip for secure communications

In a parallel development, the paper titled "Flexible entanglement-distribution network with an AlGaAs chip for secure communications" by Appas at. al. [9] underscores the importance of on-chip technologies in quantum communication networks. The use of a BRW, an on-chip, telecom-wavelength, broadband entangled photon source based on AlGaAs chip technology demonstrates scalability and reliability. The integration of flexible-grid wavelength division multiplexing techniques further en-

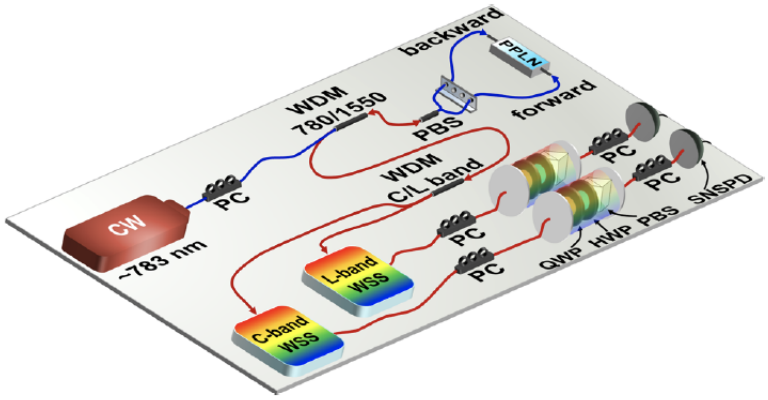


Figure 2.6: Experimental apparatus used in Alshowkan et. al. [8]. "Blue lines, polarization-maintaining fiber. Red lines, non-polarization-maintaining fiber. CW, continuous-wave tunable laser; PC, fiber polarization controller; WDM, wavelength division multiplexer; PPLN, periodically poled lithium niobate waveguide; WSS, wavelength-selective switch; PBS, polarizing beam splitter;HWP, half-wave plate;QWP, quarterwave plate; SNSPD, superconducting nanowire single-photon detector."

hances the adaptability of the entanglement distribution network. This work offers insights into reconfigurable entanglement distribution among multiple users, a key aspect of large-scale quantum communication systems.

The apparatus implemented in Appas et. al. is shown in Figure 2.7. Here, a BRW produces polarization entangled photons via type-II SPDC, in the bell state $|\Phi^+\rangle \propto |HV\rangle + |HV\rangle$. A wavelength selective switch and coarse wavelength division multiplexer (CWDM) used for spectral routing.

Adaptive bandwidth management for entanglement distribution in quantum networks

Complementing these endeavors, the paper titled "Adaptive bandwidth management for entanglement distribution in quantum networks" by Lingaraju et. al. [10] addresses the challenges associated with effective resource provisioning in quantum networks. Leveraging flexible grid technology, the paper demonstrates reconfigurable entanglement distribution in a four-user tabletop network. The scalable approach introduced for adaptive bandwidth management offers a practical pathway for establishing and managing quality-of-service guarantees in large quantum networks.

The apparatus implemented in Lingaraju et. al. is shown in Figure 2.8. Here, a periodically poled lithium niobate (PPLN) waveguide produces polarization entangled photons via type-II SPDC, in the bell state $|\Psi^-\rangle \propto |HV\rangle - |VH\rangle$. A singular

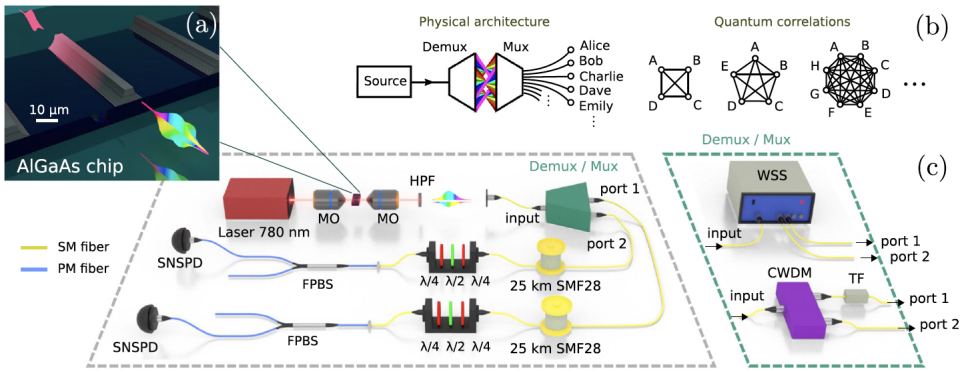


Figure 2.7: Experimental apparatus used in Appas et. al. [9]. "Source, network architecture and experimental setup. a Artist view of an AlGaAs chip under operation. An incoming pump photon is depicted on the top left corner, at the input of the waveguide. The generated entangled two-photon state is represented in the bottom right corner. b Schematic view of the entanglement network architecture. The physical structure of the network consists of an entangled photon source, a demux/mux stage and a distribution stage. The resulting quantum correlation topology takes the form of a fully connected graph of variable size (here N = 4, 5, 8), where each user node shares an entangled state with all other nodes in the network. c Experimental setup. MO: Microscope Objective. HPF: High Pass Filter. WSS: Wavelength Selective Switch. CWDM: Coarse Wavelength Division Multiplexing unit. TF: Tunable Filter. FPBS: Fibered Polarizing Beam Splitter. SNSPD: Superconducting Nanowire Single Photon Detector. SM: Single Mode fiber. PM: Polarization Maintaining fiber."

wavelength selective switch is used for spectral routing.

Relevance and Motivation

While these papers collectively advance our understanding of quantum communication networks, a critical examination reveals areas that necessitate further exploration. Specifically, the need for integrated on-chip sources capable of generating entangled photons with comprehensive spectral routing capabilities becomes apparent. In response to this research gap, our study aims to develop an on-chip two-photon source that integrates a C+L-band wavelength-selective switch and a dense wavelength-division multiplexer for efficient and scalable spectral routing within a monolithic configuration.

The BRL configuration seeks to simplify the entangled photon production process, offering a unique approach that eliminates the need for off-chip compensation or interferometry. By integrating the BRL configuration with a C+L-band wavelength-selective switch and a dense wavelength-division multiplexer, we anticipate achieving

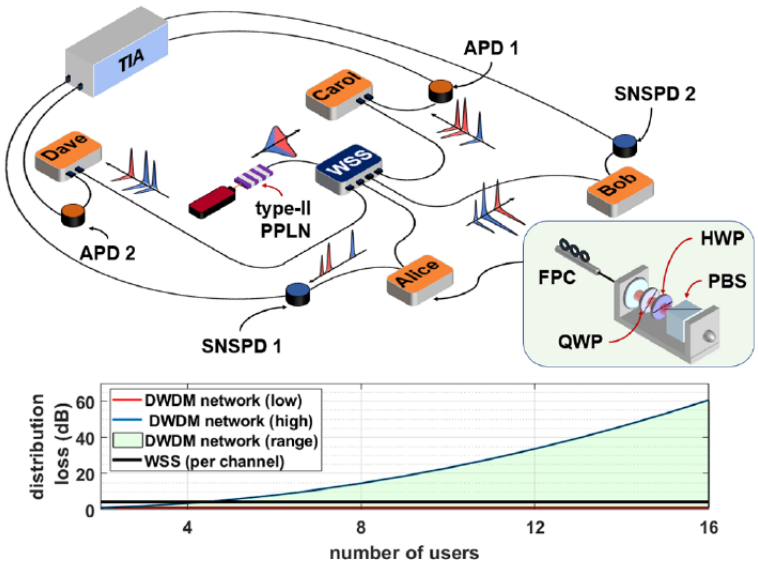


Figure 2.8: Experimental apparatus used in Lingaraju et. al. [10]. "Network testbed for adaptive entanglement distribution. The inset shows the spread and scaling of channel losses for an alternative DWDM approach compared to the WSS."

comprehensive spectral routing within a compact, on-chip setup. Our research aspires to optimize entanglement resource utilization across the C+L-band while reducing the form factor, thereby contributing to the practical implementation of quantum communication protocols, particularly those involving multiplexing.

In the subsequent sections, we delve into the specific methodologies, technologies, and experimental setups that underpin our research objectives, building upon the foundation laid by these influential papers in the field of quantum communication networks.

Chapter 3

Preliminary Tests

Before beginning the experiment, it is important to test and characterize the devices we will be using. Thus, the first half of this term has been spent testing devices.

3.1 DWDM Tests

3.1.1 Coupling Loss

The DWDM used has two input ports/channels, each with five output channels in the L-band. To test the coupling loss through each channel, a tunable CW laser was connected to the corresponding input port, and the output power was measured through the output port of said channel, using a Ge detector.

The experimental apparatus is shown in Figure 3.1 and the results are shown in Table 3.1. Although the losses vary and get as high as 64.997 %, this should still be sufficient to detect coincidences.

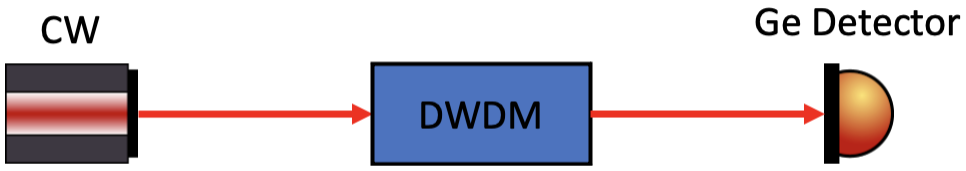


Figure 3.1: Experimental setup for coupling loss and cross-coupling loss tests.

Table 3.1: Coupling loss test results.

Input channel	Input/output wavelength (nm)	Input power (mW)	Output power (mW)	Loss (%)
A	1587.043	8.455	4.960	41.336
A	1588.725	7.760	5.470	29.510
A	1590.411	8.115	4.540	44.054
A	1592.100	7.872	4.165	47.091
A	1593.793	7.400	4.675	36.824
B	1587.884	9.125	5.021	44.975
B	1589.568	8.645	3.976	54.008
B	1591.255	8.755	4.800	45.174
B	1592.946	8.485	2.970	64.997
B	1594.641	6.137	3.891	36.598

3.1.2 Cross-Coupling Loss

To test the amount of light cross-coupling into adjacent channels, the same apparatus shown in Figure 3.1 was used. Since every subsequent output channel corresponded to opposite input ports, and since there were 10 output channels in total, the cross coupling was focused at two input wavelengths. This allowed the measurement of cross-coupling into directly neighboring channels (corresponding to a different input port), as well as neighboring channels for the same input port (which would be two output channels over on each side).

The tunable CW laser was first set at 1588.725 nm, and the output power was measured through the four neighboring channels (2 on each side), and then the same process was repeated for an input wavelength of 1592.946 nm. As seen in Table 3.2, the cross-coupling power into neighboring channels is comparable to no input power, and can thus be safely ignored.

3.2 Characterization

In order to proceed with the development of a flexible two photon source, we must first start with reliable source of entangled photons. Thus, it is crucial to identify a BRL that can be used in the experiment.

The characterization procedure follows the experimental apparatus shown in Figure 3.2. First, the BRL is probed, supplying enough current to lase, allowing for alignment of the setup. Then, connecting the detector and power supply to an automated program, the supplied (DC) current is sweeped between 0 mA and 100 mA,

Table 3.2: Cross-coupling test results.

Input wavelength (nm)	Input power (mW)	Output wavelength (nm)	Output power (mW)
1588.725	6.925	1587.043	3.825 e-3
		1587.884	3.750 e-3
		1588.725	4.165
		1589.568	3.940 e-3
		1590.411	4.025 e-3
1592.946	6.640	1591.255	4.735 e-3
		1592.100	4.740 e-3
		1592.946	3.350
		1593.793	4.730 e-3
		1594.641	4.665 e-3
None	0	None	4.057 e-3

measuring the output power at each step. This produces a luminescence-current-voltage (LIV) graph on which the lasing threshold can be identified.

Next, the output is sent to an optical spectrum analyzer (OSA), which is also connected to the automated program, and the spectrum is measured between 35 mA and 80 mA with steps of 5 mA. This allows the identification of the lasing peaks.

To ensure the devices do not overheat, an ideal BRL would have a lasing threshold around 35 mA. In order to produce SPDC around 1565 nm, an ideal BRL would also have a lasing wavelength (peak on spectrum) around 782.5 nm and be single moded. Finally, the lasing wavelength should remain single moded around the same wavelength for nearby currents so that there is room to adjust the current and temperature to find the phasematching point.

The devices tested were auxiliary cavity Fabry-Perot lasers [11]. After testing, three potential devices were found, with lasing thresholds and peak wavelengths identified in Table 3.3

Table 3.3: Characterization results of the three best devices found. Lasing threshold is identified, as well as the wavelength of the first single moded peak, along with the current at which that peak is obtained.

BRL	Lasing threshold (mA)	First peak wavelength (nm)	First peak current (mA)
BRL9_2_AS-SE-E3_W2	32.8	786.91	35
BRL9_2_AS-SE-E3_W4	34.1	788.57	35
BRL9_2_AS-SE-E3_W5	34.1	785.11	40

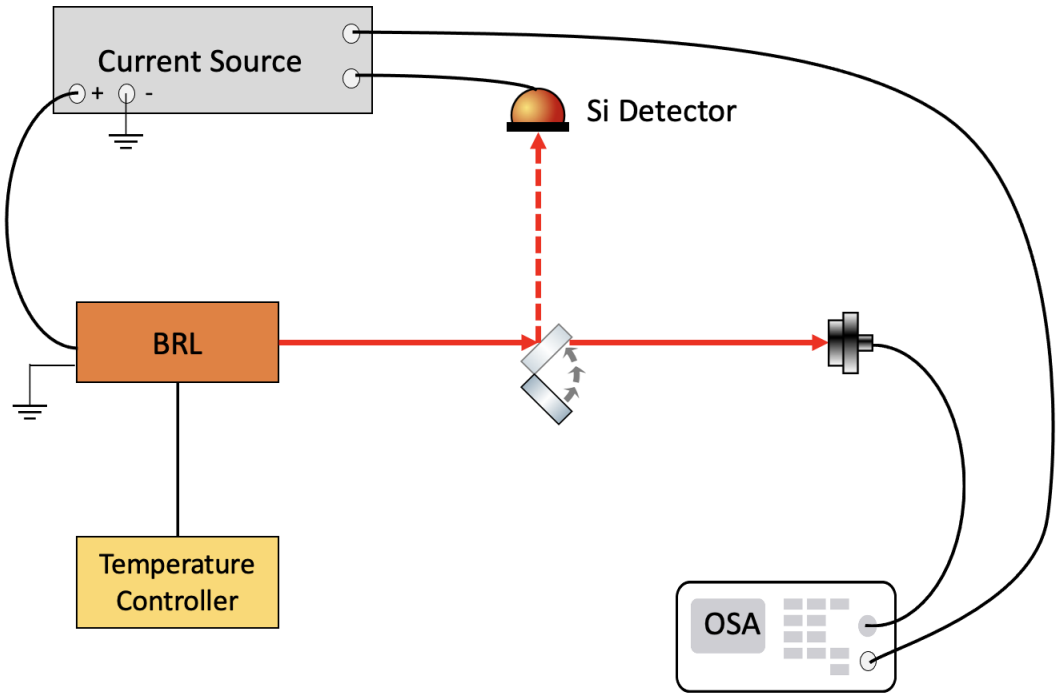


Figure 3.2: Experimental setup for BRL characterization. A flip mirror is used to flip between measuring the power and spectrum of the device when sweeping the current. The temperature controller was set at a constant 30°C.

From these results, it seems like BRL9_2_AS-SE-E3_W5 (we shall refer to as W5) is the most ideal device because of its low lasing threshold and having a single moded lasing wavelength closest to 782.5 nm. The LIV along with a sample spectrum of this device at 30°C is shown in Figure 3.3.

3.3 SHG Test

In order to produce SPDC, the phasematching (PM) point must first be identified. Since SHG occurs at a much higher rate than SPDC, it is simpler to test for SHG in order to find the PM point. SPDC and SHG are both second-order nonlinear processes that must satisfy the PM condition, so will share the same PM point.

Since SHG produces photons of double the frequency (half the wavelength) of the incident photons, one can inject a beam of 1550 nm and expect 780 nm to be produced from SHG (if PM is satisfied). The wavelength of the incident ray can be tuned until SHG in the 780 nm range is detected. The CW laser used in earlier

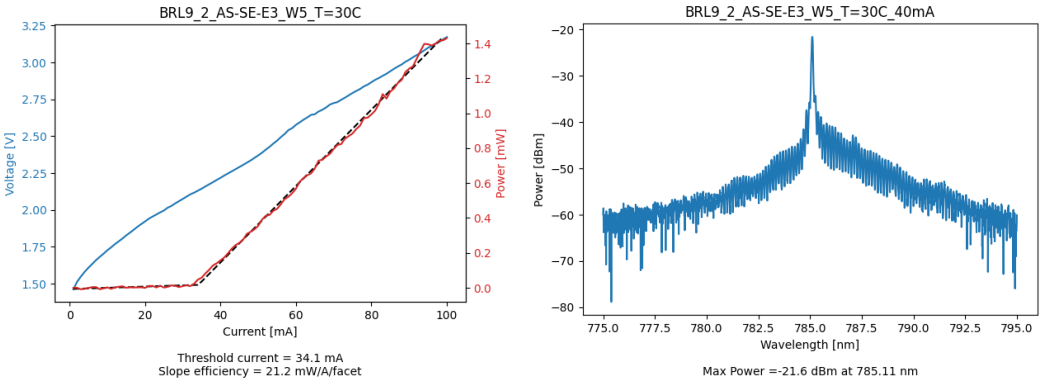


Figure 3.3: Left: LIV for the most ideal device W5 at 30°C. Right: Spectrum of W5 at 30°C and 40 mA.

tests sends exaclty (to some extent) one wavelength with a short bandwidth. This means that using the CW to test SHG would be a tedious iterative process where SHG must be tested at multiple different wavelengths until the PM point is finally reached. Thus, an alternative approach is taken.

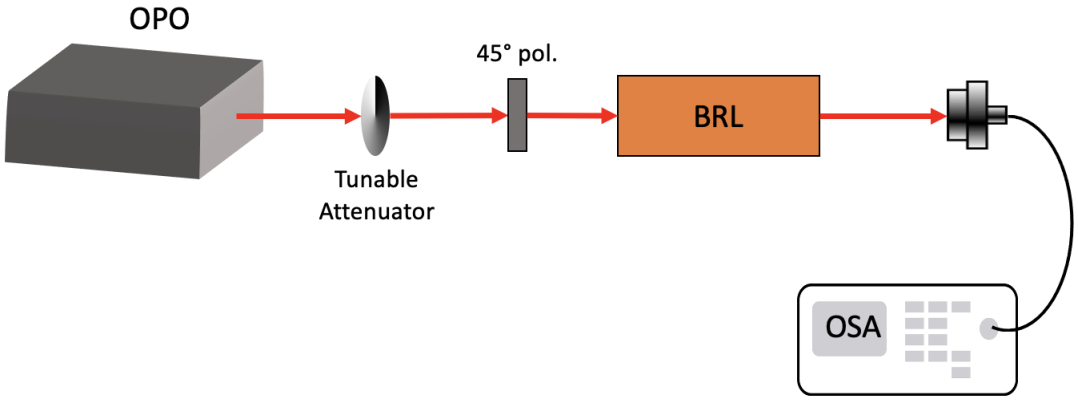


Figure 3.4: Experimental setup for testing SHG. A 1550 nm beam is attenuated and diagonally polarized before being coupled into the inactive BRL. The output through the BRL is measured on an OSA.

Figure 3.4 shows the setup used in order to test SHG. An optical parametric oscillator (OPO) is set to lase at 1550 nm. Since the OPO signal has a broad bandwidth, injecting 1550 nm from the OPO should be sufficient to 'hit' the PM point. A tunable attenuator is placed in front of the OPO to tune the input power (to ideally 1 mW to 5 mW) and ensure the BRL does not get damaged by the high power. Then the output of the BRL can be measured by an OSA which will identify points in the

780 nm range where power is measured. These specific wavelengths will be the PM points.

Since type-II phasematching involves signal and idler photons of orthogonal polarization, a 45° polarizer is also placed before the BRL to ensure the OPO input is both TE (horizontal) and TM (vertical) polarized. Type-0 and type-I involve TM and TE polarized photons, respectively.

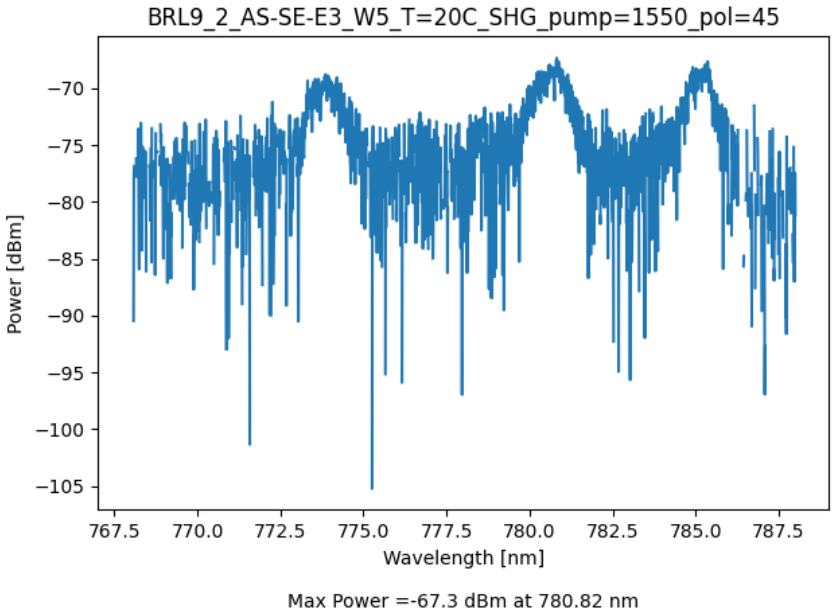


Figure 3.5: The spectrum measured from the BRL output after having injected a borad spectral range around 1550 nm. The peaks at 773.90 nm, 780.82 nm, and 785.40 nm signify type-I, type-0, and type-II pahsematching respectively.

Figure 3.5 shows the resulting spectrum when the OPO input is set at a polarization of 45°. The three peaks represent the three types of phasematching points and were at 773.90 nm, 780.82 nm, and 785.40 nm. The polarization was tuned and the change in peaks were observed in order to associate each peak with a phasematching type. The type-II phasematching point was identified to be the last peak, at 785.40 nm.

3.4 Identifying an Ideal Phasematching Point

When tuning the BRL current to lase at the PM point, it was observed that there was mode-hopping exactly at the expected PM point (see figure 3.6). That is, there were two modes, one before and one after the PM point, and as the current varied,

the dominant mode hopped from one to the other. Thus, it was not possible to tune the current such that the dominant mode remains at the PM point – these results are unstable and not desirable.

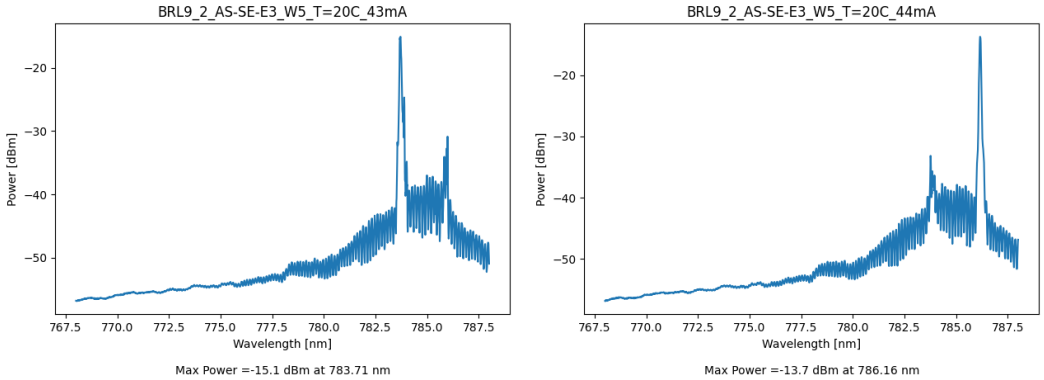


Figure 3.6: Left: BRL spectrum captured at 43 mA. Right: BRL spectrum captured at 44 mA. Mode-hopping is observed within this current range.

Thus, since the PM point and lasing wavelength varies at different rates with the BRL temperature, there could exists temperatures where the mode hopping does not occur at the PM point. Starting at 20° C, the temperature was increased/decreased in steps of 2° C. SHG was tested at each temperature, as well as the spectra at varying currents, as done previously in sections 3.2 and 3.3. Following this iterative process, 18° C was identified as a temperature where there is no mode-hopping at the PM point. At this temperature, the type-II PM point is at 784.47 nm and the behaviour was relatively single-moded.

3.5 SPDC Test

Finally, we can proceed with testing for SPDC. The BRL is lased at the PM point, and the output is sent through a 1200 nm long pass filter (LPF), as well as a 1550 band pass filter (BPF) in order to isolate the SPDC pairs from other backgrounds such as the pump photons. After filtering, a single-moded fiber splitter is used to separate the signal and idler photons, and each photon is sent towards a separate signle-photon detector. The experimental setup is shown in figure 3.7.

The time difference between the detector measurements is then measured. Assuming both photon paths are the same length, SPDC pairs should be detected at a time difference of 0. If SPDC occurs, the coincidence counts at this time difference should be higher than the average background counts, or accidentals. Thus, the

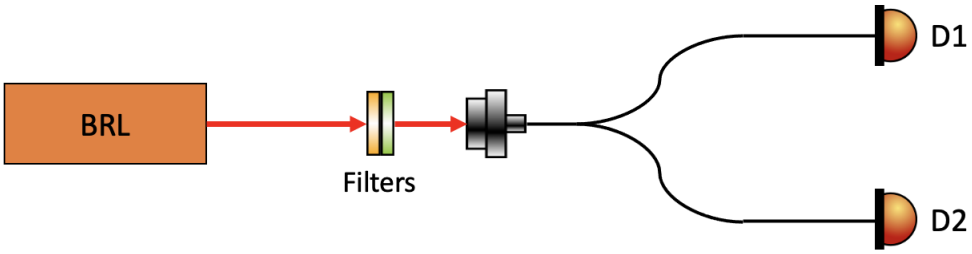


Figure 3.7: Experimental setup used to test for SPDC. The active BRL is lased at the phasematching point. The output from the BRL passes through a 1200 nm long pass filter as well as a 1550 nm band pass filter. The filtered output is coupled into a single moded fiber splitter from which the two channels are sent to two single-photon detectors D_1 and D_2.

coincidence-to-accidental ratio (CAR) is a measurement that would be proportional to the SPDC strength.

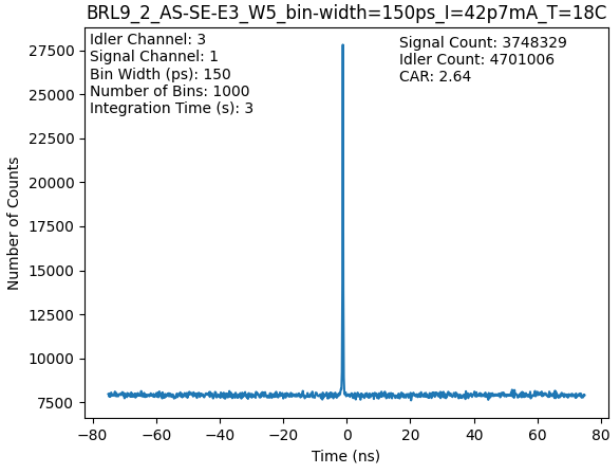


Figure 3.8: Histogram measuring the time difference between measurements of each detector. A clear peak at a time difference of 0 signifies SPDC pairs arriving at the detectors simultaneously.

A CAR greater than 1 was identified at the set current (see figure 3.8), and then the current was tuned precisely to maximize the CAR, and thus the SPDC.

Chapter 4

Experiment

Now that the relevant tools and devices have been tested, and phasematching has been confirmed and identified, we may begin the experiment.

4.1 Methods

Similar to the SPDC setup, the BRL is lased at the PM point, and the output is sent through a 1200 nm long pass filter (LPF), as well as a 1550 band pass filter (BPF) in order to isolate the SPDC pairs from other backgrounds such as the pump photons. After filtering, a single-moded fiber splitter is used to separate the signal and idler photons. One of the photons passes through our C-band WSS, and the other passes through the L-band DWDM. After passing through the filters, the coincidence counts between the detectors is measured. The experimental setup is shown in figure 4.1.

Tuning the current to optimize for maximum CAR (and thus maximum SPDC), following the same procedure in section 3.5, maximum SPDC was found to be at a current of 42.5 mA. Referring to the collected spectra for the BRL, this current was expected to produce pump photons at 785.42 nm (see figure 4.2 a).

Since the WSS is tunable and the DWDM has set wavelength channels, we label the pump wavelength as $\lambda_p = 785.42$ nm, the DWDM wavelengths as signal wavelengths λ_s, and the WSS wavelengths as the idler wavelengths λ_i. Then, using energy conservation, we can derive the the corresponding idler wavelengths for each DWDM signal wavelength:

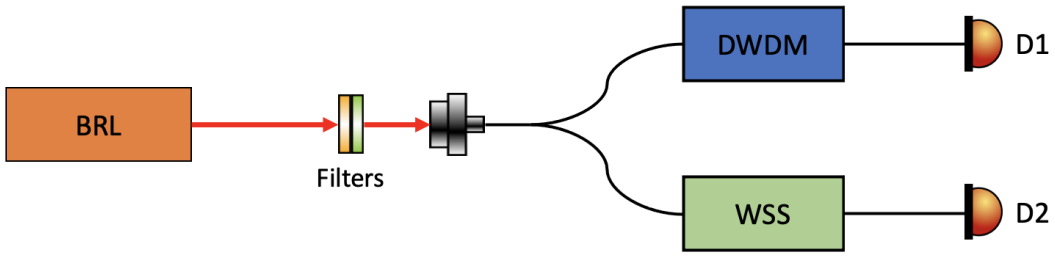


Figure 4.1: Our experimental setup used to demonstrate comprehensive spectral routing of polarization entangled photons. The BRL generates polarization entangled photon pairs through type-II SPDC, which after passing through a 1200 nm long pass filter and a 1550 nm bandpass filter, are split between a C-band WSS and an L-band DWDM. The photons are then detected at single photon detectors D_1 and D_2 and the coincidences between the two detectors are counted.

$$
\begin{aligned}
f_p &= f_s + f_i \\
\frac{c}{\lambda_p} &= \frac{c}{\lambda_s} + \frac{c}{\lambda_i} \\
\lambda_i &= \left(\frac{1}{\lambda_p} - \frac{1}{\lambda_s}\right)^{-1} = \left(\frac{\lambda_s - \lambda_p}{\lambda_p \lambda_s}\right)
\end{aligned}
$$

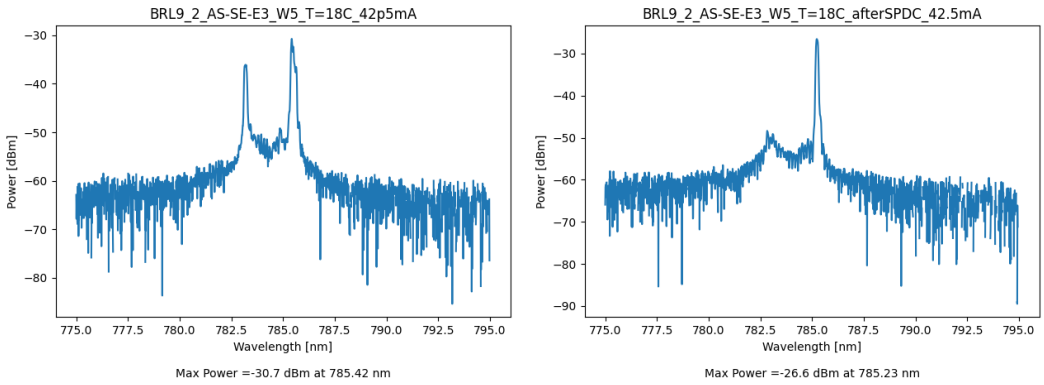
Where f_p, f_s, f_i are the pump, signal, and idler frequencies, respectively, and c is the speed of light. The set of signal wavelengths from the DWDM can be found in section 3.1 but are stated below for reference:

$$
\begin{aligned}
\lambda_s \in \{&1587.043\,\text{nm}, 1587.884\,\text{nm}, 1588.725\,\text{nm}, 1589.568\,\text{nm}, 1590.411\,\text{nm}, \\
&1591.255\,\text{nm}, 1592.100\,\text{nm}, 1592.946\,\text{nm}, 1593.793\,\text{nm}, 1594.641\,\text{nm}\}
\end{aligned} \tag{4.1}
$$

With corresponding idler wavelengths:

$$
\begin{aligned}
\lambda_i \in \{&1555.513\,\text{nm}, 1554.706\,\text{nm}, 1553.901\,\text{nm}, 1553.095\,\text{nm}, 1552.291\,\text{nm}, \\
&1551.488\,\text{nm}, 1550.686\,\text{nm}, 1549.884\,\text{nm}, 1549.083\,\text{nm}, 1548.283\,\text{nm}\}
\end{aligned}
$$

The lowest bandwidth for the DWDM channels was stated to be 141.5 THz, so for consistency and to avoid overlapping channels, the WSS was set to have a bandwidth of 141.5 THz.



(a) (b)

Figure 4.2: Left: BRL spectrum at 42.5 mA before the experiment. Right: BRL spectrum at 42.5 mA after the experiment. A clear change inmode behaviour is evident.

Measuring the coincidences for the first wavelength pair, (1587.043 nm, 1555.513 nm) resulted in no coincidence counts. This was likely due to slight changes in BRL properties due to varying temperature and current conditions, which caused a different pump wavelength to be lased than identified earlier. Thus, an iterative approach was taken, starting at a large bandwidth for the WSS and varying the WSS wavelength as well as narrowing the bandwidth until a WSS wavelength was found where coincidences were being measured.

This process narrowed down the WSS (signal) wavelength to 1553.828 nm with a bandwidth of 1.2 nm. Using the new wavelength pair, (1587.043 nm, 1553.828 nm), the current pump wavelength was derived to be 785.130 nm, and the following new idler wavelengths were identified:

$$\begin{aligned}\lambda_i \in \{&1553.828\,\text{nm}, 1553.023\,\text{nm}, 1552.219\,\text{nm}, 1551.419\,\text{nm}, 1550.613\,\text{nm},\\ &1549.812\,\text{nm}, 1549.011\,\text{nm}, 1548.211\,\text{nm}, 1547.412\,\text{nm}, 1546.613\,\text{nm}\}\end{aligned} \tag{4.2}$$

Using the sets (4.1) and (4.2), a 10x10 grid can be created to measure the CAR at each wavelength pair. The DWDM wavelength was set, and then the WSS wavelengths were varied while measuring CAR at each pair. Then the DWDM wavelength was changed to the next channel, and the process was repeated.

4.2 Expectations

The C-band WSS and the L-band DWDM will be sweeped over their entire respective ranges, measuring coincidence counts at each wavelength pair. A 2D graph will be created where the vertical axis represents the wavelength of the L-band (signal) photon, the horizontal represents the frequency of the C-band (idler) photon, and the coincidence counts are represented by the color intensity of data points.

Sweeping both these ranges and recording the coincidence counts should produce a graph similar to Figure 4.3, as demonstrated by Alshowkan et. al. [8]. A decreasing diagonal trend is expected because in order to satisfy the energy conservation condition, as the frequency of one photon increases, the frequency of the other must decrease. We will refer to these diagonal pairs as the "ideal" pairs.

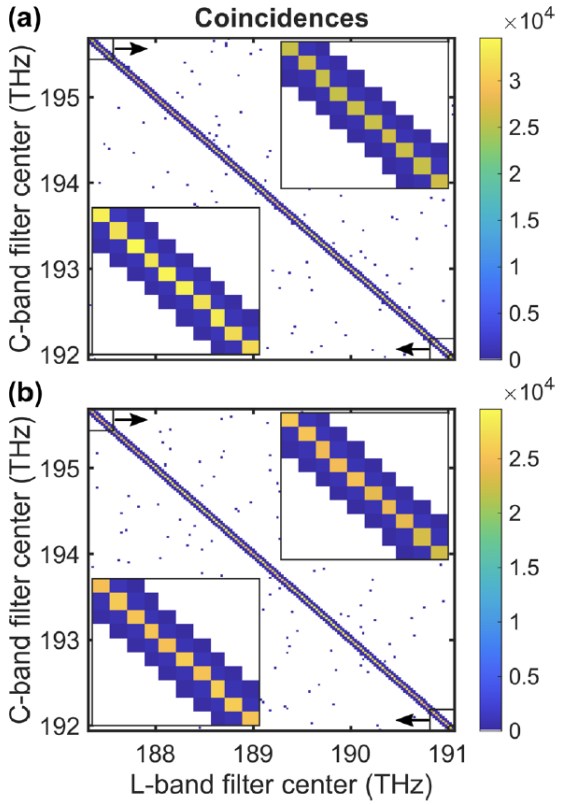


Figure 4.3: Coincidence counts plot by Alshowkan et. al. [8]. "Partial JSI measurement in the (a) forward and (b) backward directions plotted as a function of signal (C-band) and idler (L-band) channel numbers. A 1-ns coincidence window and 1-s integration time are used to obtain each point. The insets highlight the first and last 10×10 grids."

In order to gain some intuition about what is happening when the photons pass

through the filters, we can think of the SPDC photons to be in a superposition state of wavelength pairs [17]:

$$|\Psi\rangle = \int_0^\infty \int_0^\infty c_{\lambda_s,\lambda_i} |\lambda_s\rangle \otimes |\lambda_i\rangle \, d\lambda_s \, d\lambda_i \tag{4.3}$$

where the integral is over all possible wavelength pairs λ_s λ_i, and c_{λ_s,λ_i} is the probability amplitude of observing that wavelength pair. Theoretically, $c_{\lambda_s,\lambda_i} = 0$ for any λ_s, λ_i where energy conservation, $\frac{1}{\lambda_p} = \frac{1}{\lambda_s} + \frac{1}{\lambda_i}$, is not satisfied.

Passing the signal and idler photons through the DWDM and WSS then collapses the state (4.3) into the corresponding wavelength pair state. For a DWDM and WSS set at wavelengths λ_s and λ_i, passing the photons through these filters is equivalent to applying the following projector operator on the state:

$$P = (|\lambda_s\rangle \langle\lambda_s|) \otimes (|\lambda_i\rangle \langle\lambda_i|)$$

So the resulting state is (ignoring any global phase)

$$|\Psi'\rangle = \frac{P|\Psi\rangle}{|\langle\Psi| P |\Psi\rangle|^{1/2}} = |\lambda_s\rangle \otimes |\lambda_i\rangle$$

with probability

$$\langle\Psi| P |\Psi\rangle = |c_{\lambda_s,\lambda_i}|^2$$

In Zacharie Léger's PhD thesis [16] the SPDC bandwidth is found, as shown in figure 4.4. Signal/idler wavelengths are interchangeable for the horizontal axis, and the vertical axis here is proportional to the probability amplitude of that wavelength appearing in a SPDC pair. For ideal (diagonal) wavelength pairs that fall within the wavelength range [1550nm, 1600nm], we can safely assume from the plot that there will be a relatively equal probability of detecting these SPDC wavelength pairs. It is important to note though that the last few pairs, where the idler photons are $\lambda_i \in \{1549.812\,\text{nm}, 1549.011\,\text{nm}, 1548.211\,\text{nm}, 1547.412\,\text{nm}, 1546.613\,\text{nm}\}$, could have a lower probability of being detected.

4.3 Results

After measuring the CAR at all of the wavelength pairs, the heatmap in figure 4.5 was created. Points where the CAR had a distinct peak, and thus where SPDC was

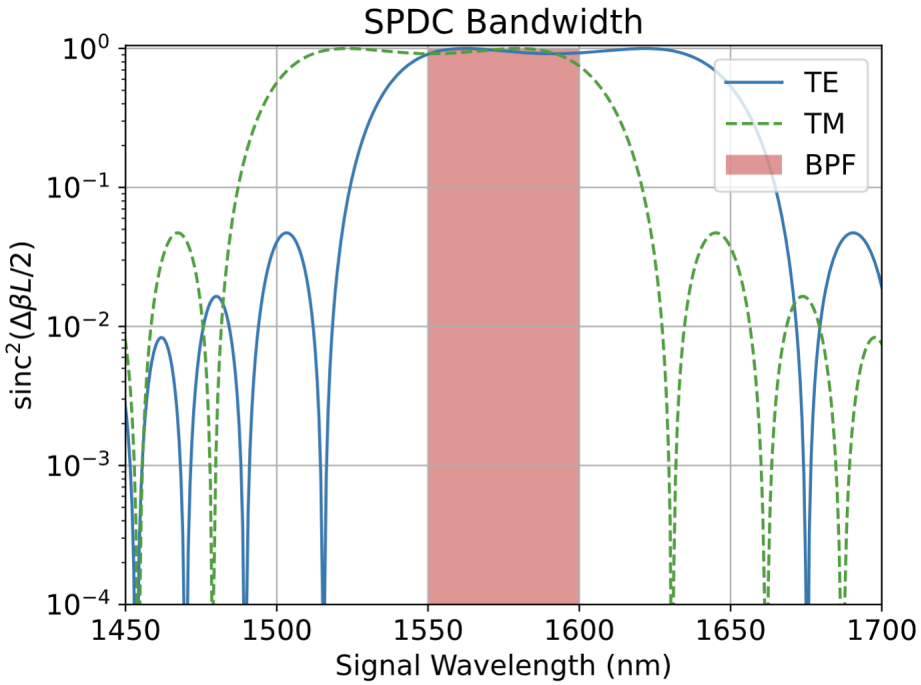


Figure 4.4: SPDC bandwidth from [16].

evident, are highlighted in red on the plot.

Note that the measured CAR depends on the chosen bin widths on the coincidences plot. All of the plots shared the same settings so the relative CAR between different wavelength pairs is what is relevant to our discussion.

4.4 Discussion

It is evident from figure 4.5 that our system exhibited the general diagonal behaviour that we expected. Although the diagonal behaviour is observed, there are rows without any significant coincidences, as well as for some rows, the CAR peaks at points away from the diagonal. It is important to discuss these anomalies.

As mentioned earlier in section 4.1, changes such as slightly varying temperature and current conditions can change BRL properties, and thus the lasing wavelength. The change in temperature as well as slight vibrations can also affect the alignment of the SPDC pairs, causing the relevant photons to drift away from the detectors, resulting in no coincidences counted. Such changes can also cause the SPDC to drift back into place. Thus, I suspect this is what may have happened when measuring CAR values for DWDM wavelength at 1589.568 nm; the SPDC photons drifted away during the measurements, and when time came to measure the CAR values at DWDM

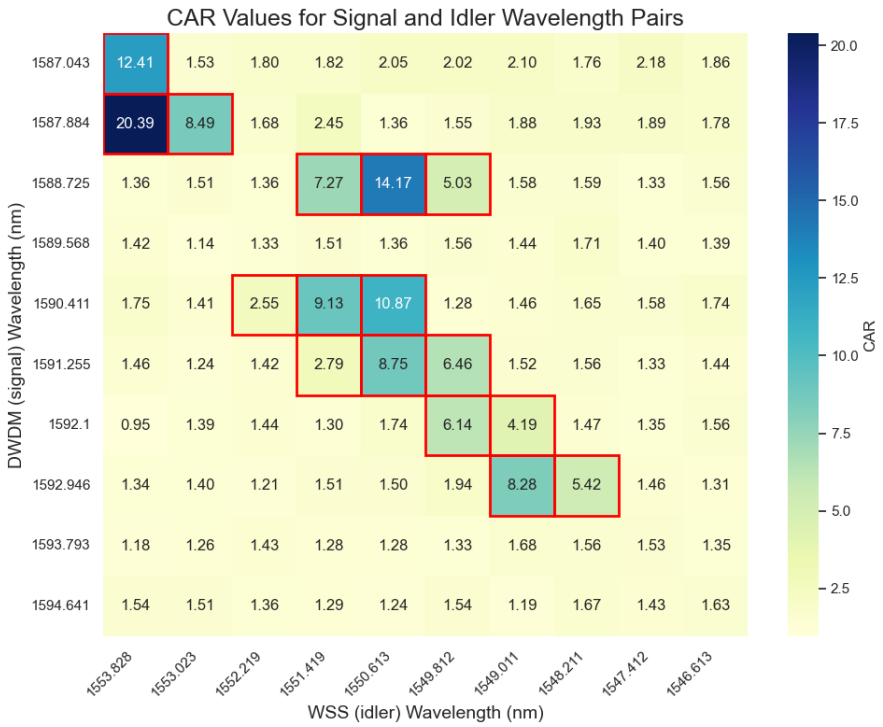


Figure 4.5: Resulting heatmap of CAR measurements between all DWDM and WSS channel pairs. Ideal CAR measurements with evident peaks are outlined in red.

wavelength 1590.411 nm, the SPDC photons had drifted back into place.

As seen on figure 4.2 (a), the spectrum of the BRL at 42.5 mA was measured to be multimoded before the experiment, despite achieving the maximum coincidence counts. This is a very unstable condition as the dominant mode could switch back and forth either completely avoiding the PM point, or switching to a different pump wavelength and thus creating SPDC pairs with different signal-idler values. This could possibly explain why certain values were shifted away from the diagonal.

Although over a short period of time variations in temperature and current conditions causes minor changes in the BRL properties, over a longer period of time these changes could add up to significant changes in the BRL properties. This is evident when comparing the BRL spectrum at 42.5 mA before the experiment in figure 4.2 (a), to after the experiment in figure 4.2 (b). Thus, over the course of the experiment, the PM point and the pump wavelength could have shifted which would explain the eventual loss of coincidences in the last two DWDM channels. This change in lasing condition could also be another reason why the ideal CAR values are shifted away from the diagonal.

Finally, although we expected only the diagonal values to have a significant CAR,

most rows have multiple points with a significant CAR value. This can be associated with the fact that the bandwidth for the WSS was 1.2 nm which would cause overlap between the WSS channels.

Despite these observed flaws, our results exhibit a diagonal coincidence pattern consistent with the energy conservation condition of SPDC. This suggests successful entanglement generation within the C+L-band spectrum. The significance of these findings lies in providing insights into the challenges and intricacies of entanglement experiments, emphasizing the importance of stability in experimental parameters. This study serves as a foundational step for future research, highlighting the need for enhanced experimental control and stability to further optimize entanglement resources for practical quantum applications.

4.5 Improving Results

To enhance the quality and reliability of the experimental results, several adjustments to the methodology could be considered. First, minimizing the potential for alignment drifting by bringing the filters, detectors, and the rest of the setup closer to the BRL output is crucial. This modification aims to reduce the impact of environmental factors such as temperature variations and vibrations, which can contribute to shifting SPDC pairs away from detectors during measurements. This adjustment would enhance the stability of the experimental setup, reducing the likelihood of data inconsistencies.

Moreover, recognizing the sensitivity of the BRL properties to time-dependent changes, there is a critical need to expedite the experimental procedure. The experiment took approximately 12 hours which, as illustrated by the spectrum at 42.5 mA before and after, highlights the potential for significant alterations in BRL properties over time. Shortening the experimental timeline would mitigate these changes and provide a more accurate representation of the entanglement properties.

Additionally, investing more time in the selection of a BRL that is not multimoded and does not mode-hop near the phasematching point would address stability concerns, minimizing the potential for fluctuations in dominant modes that can impact the SPDC pairs.

Precision in experimental parameters is paramount. Specifically, fine-tuning the bandwidth of the WSS to ensure exact diagonal values and eliminate multiple points in each row would enhance the accuracy of the results. This adjustment would reduce the overlap between WSS channels, providing a clearer representation of the expected

diagonal coincidence pattern.

4.6 Next Steps

To build on the current findings and advance the research, the next steps involve refining the experimental approach and extracting deeper insights into the entangled states within the C+L-band spectrum. With improved results that address potential sources of experimental variations, such as alignment drifting and BRL property changes, a more robust identification of ideal channel pairs exhibiting the decreasing diagonal trend on the graph will be achieved. These pairs, encapsulating frequency combinations satisfying the energy conservation condition, will serve as the focal point for further analysis.

The subsequent phase involves a meticulous quantum state tomography of each identified ideal pair. This detailed characterization aims to provide a comprehensive understanding of the polarization entangled states associated with specific frequency combinations. Calculating the fidelity for each tomographed state, by comparing it to the theoretically expected state, will quantitatively measure the closeness of the experimentally obtained states to the ideal ones. This fidelity analysis is crucial for assessing the accuracy and reliability of the generated entangled states.

To draw conclusive insights from the experiment, the calculated fidelity values will be averaged to derive an overall average fidelity. This averaged fidelity value serves as a key metric, providing a comprehensive assessment of the success and quality of the entanglement generation within the C+L-band spectrum. The insights gained from these refined analyses not only contribute to the fundamental understanding of quantum entanglement but also pave the way for the optimization of entanglement resources in practical quantum communication applications. Moving forward, this research sets the stage for further advancements in quantum photonics, ultimately contributing to the development of more efficient and reliable quantum communication protocols.

4.7 Conclusion

In conclusion, despite encountering challenges reflected in the experimental results, such as alignment drifting and fluctuations in the properties of the Bragg-Reflection Waveguide Diode Laser (BRL), we successfully observed a general diagonal behavior on the frequency graph. This behavior, while anticipated, revealed nuances and

anomalies that prompt a critical examination of our experimental approach. The identification of ideal diagonal pairs, even in the face of experimental complexities, remains a significant achievement.

The broader significance of this research lies in its potential to optimize entanglement resources within the C+L-band spectrum. The experiment's outcomes, while acknowledging areas for improvement, offer a foundational dataset for future investigations. The anomalies observed prompt a crucial reevaluation of experimental conditions, paving the way for enhanced techniques and protocols in the rapidly growing field of quantum photonics.

Beyond the immediate findings, our research propels the field forward by expanding the toolbox for quantum communication researchers. The recognition of the C+L-band spectrum as a viable resource for entanglement opens new avenues for efficient and secure quantum communication. As we navigate the challenges encountered in this experiment, we acknowledge that they serve as stepping stones for future developments in multiplexing quantum communication protocols.

In essence, this research not only achieves its immediate goals of understanding entanglement within a specific frequency range but also positions itself as a catalyst for advancements in quantum photonics. By addressing challenges and refining our approach, we contribute to a deeper understanding of quantum phenomena and set the stage for practical applications in the evolving landscape of quantum communication.

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